



Lecture 18

Previously we found the impulse response of the transfer function.

Now we would like to find the step response of a general 2nd order system.

$$\frac{Y(s)}{U(s)} = H(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$Y(s) = H(s) \cdot U(s)$$

for a step function, $U(s) = \frac{1}{s}$

$$Y(s) = H(s) \cdot \frac{1}{s}$$

$$= \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \cdot \frac{1}{s}$$

The derivation has been shown in the slide. After manipulations we find the step response as follows

$$y(t) = 1 - e^{-\zeta\omega_n t} \left(\cos \omega_d t + \frac{\zeta}{\omega_d} \sin \omega_d t \right)$$

The step response of a second order system can be characterized by several parameters.

The peak time: t_p

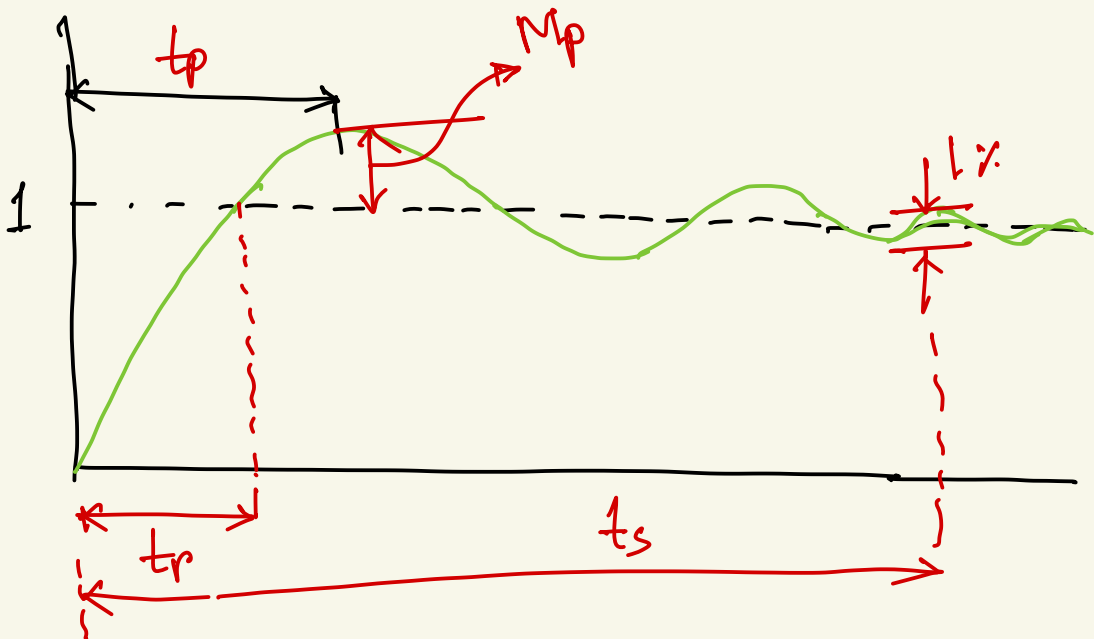
percent overshoot: M_p

Rise time: t_r

settling time: t_s

Response value at peak time
: $y_p = y(t_p)$

It can be shown below graphically.



t_p = the time at which the system reaches its peak value.

M_p = % overshoot

t_s = settling time: the time required for the system to be within 1% of the final value.

t_r = rise time: time required for the system to go from 0 to 1.

In order to find the peak times, note that the slope is zero at

the peak value. As a result, the derivative is zero at t_p .

$$y(t) = 1 - e^{-\xi \omega_n t} \left(\cos \omega_d t + \frac{\xi \omega_n}{\omega_d} \sin \omega_d t \right)$$

$$\frac{dy}{dt} = \xi \omega_n e^{-\xi \omega_n t} \left(\cos \omega_d t + \frac{\xi \omega_n}{\omega_d} \sin \omega_d t \right) - e^{-\xi \omega_n t} \left(-\omega_d \sin \omega_d t + \frac{\xi \omega_n}{\omega_d} \cdot \omega_d \cos \omega_d t \right)$$

$$= e^{-\xi \omega_n t} \cos \omega_d t (\xi \omega_n - \xi \omega_n)$$

$$+ e^{-\xi \omega_n t} \sin \omega_d t \left[\frac{\xi^2 \omega_n^2}{\omega_d} + \omega_d \right]$$

$$= e^{-\xi \omega_n t} \left(\frac{\xi^2 \omega_n^2}{\omega_d} + \omega_d \right) \sin \omega_d t$$

At peak time, this expression has to be zero.

$$\dot{y}(t_p) = 0$$

$$\Rightarrow e^{-\zeta \omega_n t_p} \left(\frac{\zeta \omega_n}{\omega_d} + \omega_d \right) \sin \omega_d t_p = 0$$

$\Rightarrow$ Cannot be zero

$$\Rightarrow \sin \omega_d t_p = 0 = \sin k\pi$$

$$\Rightarrow \omega_d t_p = k\pi \quad k = 1, 2, \dots$$

$$\Rightarrow t_p = \frac{\pi}{\omega_d} = \frac{\pi}{\omega_n \sqrt{1 - \zeta^2}}$$

